



REPUBLIC OF GUINEA

Research theme: "Optimizing synergies between forest landscape restoration and climate-adapted livestock practices in the Republic of Guinea"

Vayanga DONZO

ARIN study work

Master II student in Development

General University of Lansana Conté de Sonfonia in Conakry Email: donzovayanga@gmail.com

Phone : (00224) 620001089

1. Background and Rationale

Forest Landscape Restoration (FLR) is now recognized as one of the most effective strategies to simultaneously respond to land degradation, biodiversity loss, food insecurity and the increasing impacts of climate change. Globally, more than two billion hectares of land are considered degraded, directly affecting the livelihoods of nearly 3.2 billion people (IPBES, 2018). In response, several international initiatives, including the Bonn Challenge, the United Nations Decade on Ecosystem Restoration (2021–2030) and commitments under the Paris Agreement, are encouraging countries to develop integrated approaches that reconcile ecological restoration, rural development and climate resilience.

In sub-Saharan Africa, land degradation is a major challenge for sustainable development. Agricultural and pastoral systems, which support the livelihoods of millions of rural households, are particularly vulnerable to the combined effects of population pressure, unsustainable use of natural resources and climate change. In this context, the integration of livestock activities into landscape restoration strategies appears to be a central issue but still insufficiently explored, especially in countries where pastoral systems occupy an important place in the rural economy.

The Republic of Guinea is a particular example of these issues. Considered the "water tower of West Africa", it is home to the headwaters of several major international rivers, including the Niger, Senegal and The Gambia, and plays a key ecological role in regional hydrological regulation. However, the country's forest, agroforestry and pastoral ecosystems are increasingly degraded as a result of extensive agriculture, unsustainable logging, bush fires, population pressure and the effects of climate change. National analyses indicate that between 2000 and 2021, about 701,600 hectares of land were degraded, or nearly 2.86% of the national territory.

Among the factors of degradation identified, extensive livestock farming systems occupy an important place. The results of this study show that 80% of respondents consider overgrazing to be a major cause of land degradation, while 73% associate extensive livestock farming with increasing pressure on forest resources and 67% identify bushfires as an aggravating factor of ecological degradation. The surveys also revealed an increase in conflicts related to access to natural resources, particularly around transhumance corridors, access to pasture and occupation of rural land. However, despite these environmental impacts, livestock remains a fundamental pillar of rural livelihoods in Guinea. It contributes to food security, income generation and economic resilience in many communities. This reality highlights the existence of a major development challenge: how to reconcile the objectives of restoring forest landscapes with the needs of pastoral production and the livelihoods of rural populations?

Although many restoration initiatives are currently being implemented in the country within the framework of the National REDD+ Strategy, the Land Degradation Neutrality (LDN) targets, the Nationally Determined Contributions (NDCs) and the Bonn Challenge, the interactions between forest restoration and livestock systems are still poorly documented in the Guinean scientific literature. Mechanisms for reducing the trade-offs between ecosystem conservation, pastoral production and climate adaptation remain insufficiently studied, thus limiting the development of integrated territorial policies.

It is with this in mind that this research was undertaken. It aims to analyze synergies, trade-offs, and governance mechanisms that can foster the integration of forest landscape restoration and climate-smart livestock in the Republic of Guinea. More specifically, the study seeks to identify agro-sylvo-pastoral practices capable of restoring degraded lands, strengthening the climate resilience of rural communities and simultaneously supporting the national goals of sustainable development, food security and ecosystem conservation.

2. The objectives of this research

The objective is to analyze the synergies and trade-offs between forest landscape restoration and climate-resilient livestock practices in the Republic of Guinea, in order to identify integrated policy levers as well as institutional and technical solutions capable of reconciling environmental sustainability, food security and improved rural communities' livelihoods.

3. Research Methodology

This study adopted a mixed approach combining qualitative and quantitative methods to analyze the interactions between forest landscape restoration, climate-smart livestock practices and the Sustainable Development Goals (SDGs) in the Republic of Guinea. The analytical framework mainly focused on SDGs 2 (Zero Hunger), 13 (Climate Action) and 15 (Life on Earth), emphasizing synergies, trade-offs and territorial governance mechanisms.

The research consisted of an in-depth literature review of national environmental, pastoral and forest policies, including REDD+ strategies, Nationally Determined Contributions (NDCs), Land Degradation Neutrality (LDN) documents, as well as land and environmental legislation.

Field data collection relied on three main techniques: semi-structured one-on-one interviews conducted between August and October 2025, multi-stakeholder focus groups, and quantitative surveys using questionnaires. A total of 65 participants were involved, including local communities, herders, farmers, researchers, NGOs, environmental activists and institutional representatives.

Qualitative data were analyzed using transcription, thematic coding, content analysis, and information triangulation, while quantitative data were analyzed using simple descriptive statistics (frequencies, percentages, and comparative tables). The study also incorporated a regional comparative approach to identify best practices in land restoration and environmental governance implemented in selected West African countries.

4. Evidence-based policy-making

This research has contributed to strengthening the production of scientific and analytical knowledge to support the development of evidence-based public policies in the areas of forest restoration, sustainable livestock and climate adaptation in the Republic of Guinea.

The analyses carried out showed that an integrated territorial approach combining forest landscape restoration, climate-adapted agro-sylvo-pastoral practices and participatory governance could simultaneously generate co-benefits in environmental, climatic, economic and social terms. In particular, the study showed that some integrated practices, such as agroforestry, assisted natural regeneration (ANR), silvopastoral systems and sustainable pasture management, contribute to improving the climate resilience of rural communities, gradually restoring the ecological functions of landscapes, increasing the availability of fodder resources, improving food security and reducing some pressures on resources forests.

The results also provided a better understanding of the interactions between the Sustainable Development Goals (SDGs), in particular SDG 2 on food security, SDG 13 on climate action and SDG 15 on terrestrial ecosystem restoration.

The study showed that synergies between these SDGs depend strongly on the quality of territorial governance mechanisms, intersectoral coordination and the participation of local communities in decision-making processes.

Field surveys have highlighted several structural constraints limiting the effectiveness of current public policies, including weak coordination between environmental, agricultural and pastoral policies; the inadequacy of local consultation mechanisms; recurrent land conflicts; limited access to climate-smart innovations for vulnerable households; and the lack of inclusive territorial governance mechanisms.

In response to these challenges, the research identified several strategic orientations adapted to the Guinean context, namely: strengthening multi-stakeholder platforms, integrating agro-sylvo-pastoral approaches into national policies, securing pastoral land and land resources, developing local mediation mechanisms, supporting community-based ecological restoration practices and promoting integrated territorial models reconciling the conservation of ecosystems and rural development.

The results thus provide a useful analytical basis for public decision-makers, local governments, community-based organizations, technical and financial partners, research institutions and civil society actors engaged in environmental and climate governance in Guinea.

The study also demonstrated that participatory and inclusive territorial approaches have significant potential for replication in other regions facing similar challenges such as land degradation, pastoral pressure, and climate vulnerability in West Africa.

Overall, this research has demonstrated that the success of forest restoration and sustainable livestock policies depends less on an increase in isolated technical interventions than on the ability of local institutions and stakeholders to develop integrated and inclusive governance mechanisms, based on sound science — mechanisms that can transform territorial trade-offs into sustainable synergies between forest conservation and sustainable livestock conservation. environment, food security and climate resilience.

5. Results achieved

This study adopted a mixed approach combining qualitative and quantitative methods to analyze the interactions between forest landscape restoration and climate-adapted livestock practices in the Republic of Guinea.

This approach made it possible to triangulate the data collected from various institutional, community and technical actors involved in natural resource management, livestock and environmental policies. The study was based on: a literature review, semi-structured individual interviews, focus groups and a questionnaire survey.

The main data collection tools used were: a semi-structured interview guide, questionnaires administered to targeted stakeholders, and document summary sheets. The analysis was conducted in several locations across Guinea to cover different agroecological and socioeconomic contexts related to livestock and forest restoration.

5.2. Literature review

An in-depth literature review was conducted between July and October 2025 to analyze national policies on the environment, livestock and forest restoration; technical and institutional reports; national REDD+, NDC and LDN strategies; and scientific studies on SDGs 2, 13 and 15, as well as reports on forest restoration and pastoral development projects in Guinea and West Africa.

The main documentary sources consulted included reports from the Ministry of Environment and Sustainable Development, FAO documents, FIP-Guinea reports, UNDP publications, data from Global Forest Watch, technical reports on assisted natural regeneration (ANR), as well as documents from PNDES and the REDD+ strategy. This literature review phase identified trends in landscape degradation, dominant pastoral systems, and existing environmental governance mechanisms in Guinea.

5.2.1. Field data collection

To better understand the interactions between forest landscape restoration, livestock practices and environmental governance mechanisms in the Republic of Guinea, field data were collected in several natural regions of the country. This empirical phase made it possible to compare the documentary data with the local realities observed on the ground and to integrate the points of view of the key stakeholders involved.

The study adopted a participatory and multi-stakeholder approach to collect diverse information on the environmental, socio-economic and institutional dynamics related to agro-sylvo-pastoral systems. Primary data were collected through semi-structured individual interviews, focus groups, and questionnaires.

5.2.2. Semi-structured individual interviews

Semi-structured one-on-one interviews were one of the main qualitative tools used in this study. They took place from August 1 to 28, 2025 in the cities of Conakry, Kindia, Labé and Nzérékoré.

These locations were chosen because of their ecological, pastoral and institutional importance in addressing issues of forest restoration and livestock management systems in Guinea. These areas also represent different agroecological contexts within the country.

A total of ten (10) individual interviews were conducted with key stakeholders involved in natural resource management, livestock, rural development and environmental policy.

(a) Scientific justification for semi-structured interviews

The use of semi-structured interviews provided in-depth information on: local livestock practices, forest degradation dynamics, land use conflicts, ecological restoration initiatives, perceptions of climate change, and territorial governance mechanisms.

The semi-structured approach was chosen to allow respondents to freely express their experiences, constraints and perceptions while maintaining thematic coherence in the discussions.

(b) Criteria for Selecting Participants

The participants were selected according to a reasoned method (targeted qualitative sampling) based on several criteria, mainly their professional experience, their level of involvement in natural resource management, their knowledge of pastoral systems, their technical and scientific expertise, their institutional role in environmental and agricultural policies, as well as their community involvement in local restoration initiatives. This selection aimed to ensure a diversity of experiences and perspectives to better understand local realities and institutional challenges related to the interactions between livestock and forest restoration.

(c) Classes of Respondents

The interviews involved several categories of actors directly involved in the issues studied.

Table 1: Categories of participants interviewed during the interviews

Categories of stakeholders	Number
Members of local communities	2
Women's Economic Groups	2
Pastoralists	2
Small family farmers	2
State sectoral directorates	2

Categories of stakeholders	Number
Total	10

Members of local communities and small-scale rural producers were interviewed to document changes in forest landscapes, agricultural and pastoral practices, as well as land conflicts related to access to natural resources. In addition, women's economic groups provided important information on the exploitation of non-timber forest products (NTFP), local income diversification strategies, the effects of climate change on rural economic activities, and the role of women in community restoration initiatives.

However, pastoralists were interviewed about transhumance systems, pasture management practices, difficulties in accessing pastoral corridors, the effects of overgrazing, and tensions between pastoralists and farmers.

Finally, representatives of sectoral government departments provided information on public environmental policies, forest restoration programmes, territorial governance frameworks and institutional mechanisms for managing conflicts related to natural resources.

(d) **Main topics discussed during the interviews**

The semi-structured interviews were organized around several major themes, namely:

1. Traditional pastoral practices

Discussions focused on transhumance practices, large livestock systems, pastoral forest fires, local grazing management techniques and difficulties in accessing fodder resources.

The target audience reported that increasing pressure on land and lack of pastoral planning had contributed to the spread of overgrazing in several parts of the country.

2. Land Use Conflicts

The interviews identified several types of conflicts, such as conflicts between farmers and herders, conflicts related to the use of transhumance corridors, tensions around forest restoration areas, and community land conflicts.

Stakeholders interviewed pointed to the gradual reduction of grazing and unplanned agricultural expansion as major sources of tension in rural areas. **Local forest restoration initiatives**

Participants mentioned several ecological restoration initiatives observed in the areas studied: community reforestation, agroforestry, assisted natural regeneration (ANR), gallery forest restoration, and community-based natural resource management.

The interviews showed that participatory approaches involving local communities promoted greater ownership of restoration efforts.

3. Perceptions of climate change

The majority of participants — 88% — reported increasingly erratic seasons, more frequent droughts, degraded pastures, declining water resources, and reduced soil fertility.

These changes have been seen as simultaneously exacerbating food insecurity, land-use conflicts, and the vulnerability of pastoral systems.

4. Synergies and trade-offs between SDGs 2, 13 and 15

One of the central objectives of this research was to analyze the interactions between forest landscape restoration practices and livestock systems in Guinea through the lens of the Sustainable Development Goals (SDGs), including: SDG 2: Zero hunger, SDG 13: Climate action, and SDG 15: Life on land.

Analysis of data collected through interviews, focus groups, and surveys showed that the relationships between livestock, forest restoration, and climate resilience were not limited to isolated impacts, but constituted a complex system of synergies, trade-offs, and territorial governance mechanisms.

This approach went beyond a simple description of practices to examine how some interventions simultaneously contributed to several Sustainable Development Goals, while others generated social, economic or ecological tensions.

4.1. Mapping the interactions between the SDGs

Table 2: Presents a mapping of interactions between the SDGs

Observed practices	SDG 2	SDG 3	SDG 15	Type of interaction
Assisted Natural Regeneration (ANR)	Drilling availability	Carbon sequestration	Soil restoration and biodiversity	Strong synergy
Agroforestry	Agricultural productivity and livestock	Climate adaptation	Forest restoration	Strong synergy

Rotational pastures	Pastoral improvement	Reducing land degradation	Ecosystem protection	Moderate synergy
Unregulated extensive livestock farming	Immediate productivity	Emissions and climate vulnerability	Deforestation and overgrazing	Negative compromised
Semi-intensive agriculture	Reducing pressure on forests	Better climate management	Limits of pastoral expansion	Synergy with social risk
Reforestation carried out without consultation	Environmental protection	Carbon sequestration	Forest Conservation	Social compromised

Qualitative and quantitative analyses revealed that the strongest synergies occurred when environmental interventions were integrated with the economic needs of rural communities.

4.2. Synergy 1: Assisted natural regeneration (ANR), food security and climate resilience

The study showed that assisted natural regeneration (ANR) was one of the practices with the strongest positive links between SDGs 2, 13 and 15.

In the study areas of Kindia, Labé and Nzérékoré, several target populations reported that natural tree restoration in agropastoral areas improved soil fertility, reduced erosion, increased fodder availability during the dry season, improved soil moisture and restored some local plant species.

According to questionnaire data, 71% of participants felt that RV directly improved the availability of forage for livestock, 68% reported improved soil fertility, and 64% associated RV with greater resilience to recurrent droughts¹.

Table 3: Presents Synergy 1 between Assisted Natural Regeneration (ANR), Food Security and Climate Resilience

Observed effects of RNA	Percentage of respondents
Improved forage	71 %
Improving soil fertility	68 %
Erosion Reduction	59 %
Increased climate resilience	64%

This practice has thus simultaneously contributed to food and pastoral security (SDG 2), climate adaptation and carbon sequestration (SDG 13), as well as the restoration of terrestrial ecosystems (SDG 15).

¹ Analysis of the synergy of women's economic groups, 2025 Survey Questionnaires

The focus groups also showed that regenerated trees play an important role in reducing local temperatures and protecting pastures from prolonged droughts.

4.3. Synergy 2: Agroforestry and integration of agriculture, livestock and forestry

The analyses revealed that agroforestry systems represent a particularly promising approach to reconciling agricultural production, livestock and forest restoration.

In several areas studied, including in Kankan and N'zérékoré, farmers interviewed indicated that the integration of trees into agricultural and pastoral systems has improved the availability of fodder resources, increased farm incomes, reduced pressure on natural forests, and improved soil conservation.

Forage and soil improvement species cited included locust, shea, leucaena, gliricides and some local fruit species.

According to the data collected, 73% of farmers surveyed said that agroforestry had improved agricultural productivity, 61% observed a reduction in pressure on forest resources, and 58% reported an improvement in animal feed during the dry season.

Table 4: Synergy 2: Agroforestry and integration of agriculture, livestock and forestry

Contributions of agroforestry	Percentage
Improving agricultural productivity	73 %
Reducing pressure on forests	61 %
Increased forage availability	58 %
Building climate resilience	66%

These results show that agroforestry systems generate co-benefits in the areas of food security, climate adaptation and biodiversity conservation.

4.4. Trade-off 1: Pastoral productivity versus ecosystem conservation

The study also identified several significant trade-offs between the Sustainable Development Goals. In some pastoral areas of Labé and Kankan, the expansion of intensive livestock activities has led to the degradation of pastures, the decline of gallery forests, an increase in bush fires and soil erosion. Respondents indicated that pastoral pressure was increasing sharply around forest restoration areas due to reduced available space, population growth, and lack of land-use planning. According to the surveys, 76% of respondents identified overgrazing as one of the main factors contributing to land degradation, 69% reported that conflicts between farmers and herders had increased in recent years, and 54% felt that some reforestation initiatives temporarily restricted access to pasture.

Table 5: Trade-off 1: Pastoral productivity versus ecosystem conservation

Key trade-offs identified	Percentage
Overgrazing and soil degradation	76 %
Increase in land conflicts	69 %
Restrictions on access to pasture	54 %
Pressure on gallery forests	63 %

These results show that some conservation policies can create social tensions when they do not sufficiently take into account the pastoral needs of local communities.

4.5. Trade-off 2: Semi-intensive livestock farming and economic exclusion of vulnerable households

The study also showed that semi-intensive livestock systems can have positive environmental effects while creating socio-economic inequalities. Several institutional and producer stakeholders reported that barns, improved fodder crops, silvopastoral systems and livestock breeding helped to reduce pressure on forest areas and natural pastures. However, these systems required significant financial investments, access to inputs, pastoral infrastructure and technical support. According to the data collected, 62% of smallholders reported that they did not have the resources to adopt these practices, 57% felt that technical support mechanisms remained insufficient, and 48% indicated that rural women had limited access to agricultural and pastoral finance. These results reveal a significant trade-off between environmental efficiency, social inclusion and economic equity. As a result, some climate-smart innovations risk excluding the poorest households if they are not accompanied by inclusive financing mechanisms.

4.6. Governance of synergies and trade-offs

The analysis of qualitative data showed that territorial governance plays a central role in managing synergies and trade-offs between the SDGs. The main challenges identified included weak intersectoral coordination, insufficient local consultation mechanisms, land conflicts, weak pastoral management systems, and a lack of community integration in some environmental initiatives.

Participants strongly emphasized the need to develop co-management mechanisms, multi-stakeholder platforms, local pastoral land management plans, integrated agriculture-livestock-forestry approaches, and participatory governance systems.

According to the surveys, 72% of respondents believed that community participation improved the success of restoration projects, 67% stressed the importance of better coordination between environmental and livestock services, and 59% recommended the establishment of local land mediation mechanisms.

Overall, the results show that the interactions between SDGs 2, 13 and 15 are highly dependent on institutional capacities, territorial governance mechanisms and the level of inclusion of local communities in decision-making processes.

5.4.2. Focus Groups

Five (5) focus groups were held between 5 and 14 September 2025 in the cities of Conakry, Kindia, Momou and Kankan.

The focus groups brought together researchers, consultants, academics, environmental policy specialists, NGO representatives and stakeholders involved in the livestock, forest restoration and territorial governance sectors. This diversity of actors made it possible to compare scientific, institutional and community perspectives in order to better understand the complex dynamics that shape the interactions between pastoral systems and forest landscapes in Guinea.

Table 6: Distribution of Focus Groups

Rental	Number of Focus Groups	Participants by category
Conakry	2	16
Kindia	1	8
Mom	1	7
Kankan	1	9
Total	5	40

Focus groups served as a key tool for qualitative data collection in this study, facilitating interactive discussions among various actors on issues related to forest restoration, climate-smart livestock and the Sustainable Development Goals (SDGs). The group discussions were organized according to a participatory and semi-structured approach, facilitating the exchange of local experiences, scientific knowledge and institutional practices.

Discussions held in Conakry focused mainly on Guinea's public environmental policies, governance mechanisms and international commitments on climate, biodiversity and land restoration. Participants stressed that national policies often remain sector-specific and insufficiently coordinated in the areas of environment, agriculture and livestock. The discussions also highlighted the challenges related to operationalizing synergies between SDGs 2, 13 and 15 in local territorial policies.

In contrast, in Kindia and Mamou, discussions focused more on traditional pastoral systems, land-use conflicts, and the impacts of overgrazing on gallery forests and community rangelands. Several participants noted that the gradual reduction of pastoral land, combined with population growth and agricultural expansion, had increased tensions between farmers and herders.

Participants also discussed the effects of pastoral bush fires on forest regeneration and local biodiversity. However, the focus groups held in Kankan allowed for more in-depth discussions on pilot projects integrating agriculture, livestock and forestry in Guinea's savannah regions.

The target populations highlighted several practices considered innovative and adapted to the climate, including controlled pasture rotation, improvement of fodder crops, silvopastoral systems, assisted natural regeneration (ANR) and the integration of fodder trees in pastoral areas.

Discussions revealed that these approaches help reduce pressure on forest resources while improving pastoral productivity and climate resilience of rural households.

The panel discussions also identified several significant synergies between the SDGs. Participants agreed that agroforestry practices and RNA simultaneously contribute to the restoration of terrestrial ecosystems (SDG 15), improved food and pastoral security (SDG 2), and climate change adaptation and mitigation (SDG 13).

In this regard, several compromises were also put forward. Respondents noted that some reforestation policies sometimes exclude pastoralists from restored areas, temporarily limiting access to pasture and creating social tensions.

Other participants noted that semi-intensive livestock systems, although more environmentally sustainable, required significant financial investments that were difficult to access for the poorest rural households.

The focus groups also provided an opportunity to assess existing territorial governance mechanisms. Participants highlighted the weakness of intersectoral coordination, the inadequacy of local land management systems, the lack of community involvement in some restoration initiatives, as well as the need to strengthen mechanisms for co-management of natural resources.

Finally, there was widespread recognition among target populations that participatory approaches involving local communities, administrative authorities, technical services and civil society organizations have yielded better results in ecological restoration and the prevention of land-use conflicts.

Overall, the focus groups provided critical qualitative data that provided a better understanding of the interactions between livestock, forest restoration, environmental governance, and rural development in Guinea's different agroecological contexts.

5.4.3. Questionnaire Survey

As part of this research, a quantitative survey was carried out from 2 to 22 October 2025 in the cities of Conakry, Dubréka, Kindia and Labé.

This quantitative phase aimed to complement qualitative data from individual interviews and focus groups, to measure perceptions, trends and interactions between livestock systems, forest restoration practices and the Sustainable Development Goals (SDGs).

The survey particularly focused on the linkages between food security and pastoral systems (SDG 2), climate adaptation and resilience (SDG 13), and restoration of terrestrial ecosystems and biodiversity (SDG 15).

The quantitative approach made it possible to identify the dominant trends observed by institutional and non-institutional actors involved in natural resource management and environmental policies in Guinea.

a) Scientific justification for the quantitative survey

The use of a questionnaire quantified stakeholder perceptions, compared levels of understanding of environmental issues, assessed existing climate-smart practices, identified trade-offs and synergies between the SDGs, and measured levels of governance and institutional coordination.

This method has also facilitated the production of useful statistical data to enhance the analytical credibility of the research and to support qualitative results from other data collection methods.

b) Selection of participants and sampling criteria

A total of 15 questionnaires were administered to key stakeholders from various institutional and community categories.

Respondents were selected based on their environmental expertise, involvement in agricultural and pastoral policies, participation in forest restoration initiatives, role in local governance, and experience on climate and sustainable development issues.

The sampling was designed to ensure that a wide range of stakeholders had operational knowledge of the issues being studied.

(c) Distribution of respondents Table 7: Distribution of Respondents

Category of actors	Number	Percentage
Environmental NGOs	4	26,7 %
Civil Society	3	20 %
Environmental activists	3	20 %
Department of the Environment	3	20 %
Department of Agriculture	2	13,3 %
Total	15	100

Environmental NGOs and civil society organizations mainly provided information on landscape restoration projects, community practices, participatory governance mechanisms, and conflicts over natural resources.

In addition, ministry representatives provided information on public policies, forest restoration programs, climate adaptation strategies, and pastoral management systems.

In contrast, environmental activists have helped document anthropogenic pressures on ecosystems, local land degradation dynamics, and challenges related to the implementation of environmental policies.

(d) Main topics covered in the questionnaires

The questionnaires were structured around five main analytical themes:

1. The environmental impacts of livestock farming practices

Participants were asked about the effects of overgrazing, deforestation related to intensive livestock farming, pastoral bush fires, pasture degradation and land-use conflicts.

The results show that 80% of respondents considered overgrazing to be a major driver of land degradation, 73% believed that extensive livestock farming contributed to pressure on forest resources, and 67% identified pastoral bushfires as a major cause of biodiversity loss.

Table 8: Environmental impacts of livestock practices

Identified environmental impacts	Percentage
Overgrazing and soil degradation	80 %
Pressure on forest resources	73 %
Recurrent bush fires	67 %
Decline of natural pastures	61 %

These results confirm that extensive and unregulated pastoral systems are a major driver of environmental degradation in several regions of the country.

2. Climate-smart practices observed

The survey also identified key climate practices that have already been tested and are perceived to be effective by the stakeholders interviewed.

The most frequently cited practices were agroforestry, assisted natural regeneration (ANR), rotational grazing, silvopastoral systems and forage crop improvement.

According to the data collected, 74% of respondents considered agroforestry to be an effective solution to reduce pressure on forests, 69% indicated that ANR improved the climate resilience of rural farms, and 63% said that silvopastoral systems contributed to better livestock feeding in times of drought.

Table 9: Climate-smart practices observed

Climate-smart practices	Percentage of favourable responses
Agroforestry	74 %
Assisted natural regeneration	69 %
Silvopastoral systems	63 %
Rotational pastures	58 %
Improved forage crops	54 %

These results showed that several integrated solutions are already known locally and perceived as compatible with forest landscape restoration objectives.

3. Forest restoration policies

Participants also assessed forest restoration policies and programmes implemented in Guinea. The results showed that 72% of respondents considered that restoration initiatives lacked intersectoral coordination, 65% felt that local communities were insufficiently involved in projects and 59% considered restoration policies insufficiently adapted to pastoral realities.

Table 10: Forest restoration policies

Evaluation of restoration policies	Percentage
Lack of institutional coordination	72 %
Low community involvement	65 %
Insufficient integration of livestock farming	59 %
Difficulties of local application	63 %

These data highlight the need for a more integrated approach between forest restoration, pastoral management and territorial governance.

4. Local governance mechanisms

The questionnaires also explored local mechanisms of environmental governance. The participants highlighted the lack of consultation between actors, the inadequacies of land tenure systems and the weakness of conflict prevention mechanisms.

According to the data collected, 70% of respondents felt that land conflicts were increasing in rural areas, 67% recommended the development of local mediation mechanisms and 74% considered it necessary to strengthen participatory approaches in environmental projects.

5. Interactions between SDGs 2, 13 and 15

One of the major axes of the questionnaire concerned the analysis of synergies and trade-offs between the SDGs.

The results showed that 76% of respondents considered that forest restoration could sustainably improve food security through improved soils and pastures, 71% believed that agroforestry systems simultaneously enhanced climate adaptation and agricultural productivity, and 62% reported that some conservation measures temporarily limited access to pastoral resources.

Table 11: Interactions between SDGs 2, 13 and 15

Interactions between SDGs	Percentage
Synergies between catering and food security	76 %
Agroforestry and climate resilience	71 %
Retention Restrictions	62 %
Need for integrated governance	79 %

These results show that interactions between SDGs can simultaneously produce environmental benefits and socio-economic tensions when they are not accompanied by inclusive governance mechanisms.

(e) Analytical contribution of the quantitative survey

Overall, the questionnaire survey strengthened the validity of the qualitative results and identified dominant trends related to the interactions between livestock and forest restoration, quantified stakeholder perceptions, better understood synergies and trade-offs between SDGs 2, 13 and 15, and highlighted institutional limitations and integrated governance needs.

The results obtained confirm that climate-smart approaches based on agriculture-livestock-forestry integration offer significant opportunities to reconcile food security, climate resilience and ecosystem restoration in Guinea.

5. Overall sample size

The study mobilized a global sample of sixty-five (65) participants divided between the different qualitative and quantitative collection methods. The combination of individual interviews, focus groups and questionnaires made it possible to obtain a diversity of stakeholder profiles and points of view on the interactions between forest landscape restoration, livestock practices and environmental governance in the Republic of Guinea.

This methodological diversity has contributed to strengthening the scientific validity of the results through a triangulation of data from local communities, institutional actors, researchers, civil society organizations and rural and environmental development actors.

The sample was selected using a reasoned and participatory approach, focusing on actors directly involved in natural resource management, pastoral systems, environmental policies, forest restoration initiatives and territorial governance mechanisms.

Table 12: Overall Sample Distribution

Collection method	Number of participants	Percentage
Semi-structured individual interviews	10	15,4 %
Questionnaire surveys	15	23,1 %
Focus groups	40	61,5 %
Grand Total	65	100 %

Focus groups accounted for the largest proportion of participants (61.5%), which is explained by the desire to foster in-depth collective exchanges on the synergies and trade-offs between SDGs 2, 13 and 15.

On the other hand, the questionnaires made it possible to strengthen the statistical analysis of the perceptions and practices observed, while the individual interviews provided detailed information on local and institutional realities.

5.5. Data analysis procedures

The data analysis was based on an integrated approach combining qualitative and quantitative techniques to ensure a better understanding of the interactions between livestock, forest restoration, climate resilience and environmental governance.

The main objective of this analytical phase was to identify synergies and trade-offs between the SDGs (i), assess territorial dynamics (ii), analyse governance mechanisms (iii), measure stakeholder perceptions (iv) and identify the most relevant climate-smart practices (v).

5.5.1. Analysis of qualitative data

The qualitative data from the individual interviews and focus groups were processed according to several complementary methodological steps.

a) Data transcription and organization

All the interviews and collective discussions were transcribed manually in order to preserve the content of the exchanges, the discursive nuances, the perceptions of the actors and the local experiences.

The data were then organized by locality, category of actors, research theme and type of interaction between SDGs.

(b) Thematic codification

A thematic codification was carried out in order to identify the main axes of analysis recurrent in the participants' speeches.

The main thematic categories selected concerned extensive livestock farming practices, agro-sylvo-pastoral systems, land use conflicts, forest restoration mechanisms, climate adaptation strategies, environmental governance mechanisms and synergies and trade-offs between SDGs 2, 13 and 15.

This approach made it possible to identify the dominant trends as well as the divergences between the different groups of actors interviewed.

c) Content analysis

The content analysis made it possible to interpret the qualitative data in order to understand the social representations of climate change, the perceptions of environmental degradation, the local logics of natural resource management and the mechanisms of cooperation or conflict between actors.

The testimonies collected also illustrated several trade-offs between ecosystem conservation, food security, access to pastoral resources and socio-economic equity.

(d) Triangulation of information

Data from the different collection methods were compared and cross-referenced to increase the reliability of the results.

The triangulation made it possible to verify the coherence between the qualitative discourses and the statistical results, to compare the institutional data with local realities and to identify the convergences and contradictions between actors.

This methodological approach helped to reduce interpretation biases and improve the analytical robustness of the study.

5.5.2. Analysis of quantitative data

Quantitative data from the questionnaires were processed using simple descriptive statistics to produce a synthetic reading of the trends observed.

The analyses focused on perceptions of the environmental impacts of livestock, climate-smart practices, restoration policies, land conflicts, governance mechanisms and interactions between SDGs.

Statistical treatments included frequencies, percentages, comparative tables and distribution tables.

Table 13: Main Statistical Trends Observed

Theme analysed	Percentage of respondents
Overgrazing is a major cause of land degradation	80 %
Agroforestry systems improve climate resilience	71 %
Restoration policies lack coordination	72 %
Land conflicts are on the rise in rural areas	70 %
Participatory approaches improve restoration	74 %
Restoration programmes sometimes limit pastoral access	62 %

These results show that the environmental issues studied are strongly linked to issues of territorial governance, pastoral management and the integration of sectoral policies.

5.5.3. Analysis of the interactions between the SDGs

Particular attention was paid to the analysis of the interactions between SDG 2: food security, SDG 13: climate action and SDG 15: restoration of terrestrial ecosystems.

The study showed that some practices produced important synergies between these objectives, including: agroforestry, assisted natural regeneration, silvopastoral systems and sustainable pasture management.

However, several trade-offs were also identified, including tensions between forest conservation and pastoral access and economic difficulties in adopting semi-intensive systems and inequalities in access to resources and finance.

5.6. Ethical Protocols

This research respected the ethical principles applicable to studies in the social, environmental and territorial sciences.

Particular attention was paid to respect for participants, data confidentiality, free and informed consent and the protection of sensitive information.

Before each interview, focus group or questionnaire, the objectives of the research were clearly explained, the participants were informed about the scientific use of the data, their voluntary consent was obtained and their right to withdraw at any time was guaranteed.

Participants were also informed that no nominative data would be published, the results would be presented in an aggregated manner and the citations used in the study would be anonymised.

This approach has made it possible to establish a climate of trust favourable to the quality of the exchanges and the reliability of the information collected.

5.7. Statistical summary of collection activities Table 14: Overview of field activities

Activity	Period	Locations	Number of participants
Individual interviews	August 1–28, 2025	Conakry, Kindia, Labé, Nzérékoré	10
Focus groups	05 – 14 September	Conakry, Kindia, Mamou, Kankan	40

Activity	Period	Locations	Number of participants
	2025		
Questionnaires	02 – 22 October 2025	Conakry, Dubréka, Kindia, Elba	15
Grand Total			65

Table 15: Distribution of Participants by Category

Category of actors	Workforce	Percentage
Local communities and producers	24	36,9 %
Public institutions	12	18,5 %
NGOs and civil society	11	16,9 %
Researchers and academics	18	27,7 %
Total	65	100 %

Local communities and producers were the most represented category, reflecting the importance given to territorial realities and local practices in this research.

Table 16: Main themes raised by participants

Main themes	Frequency of occurrence
Forest and land degradation	88 %
Overgrazing and land conflicts	76 %
Agroforestry and assisted natural regeneration	69 %
Environmental Governance	64 %
Synergies between SDGs 2, 13 and 15	71 %
Climate change and rural resilience	67

The results show that the most frequently mentioned issues concern accelerated landscape degradation, land use conflicts, governance inadequacies and the need for integrated approaches reconciling forest restoration, food security and climate resilience.

Overall, this statistical analysis confirms that the interactions between livestock, forest restoration and environmental governance are a central issue for the sustainability of socio-ecological systems in the Republic of Guinea.

6. Major result obtained from this research

6.1. Result 1: Extensive livestock farming practices are a major factor in ecological degradation and pressure on forest landscapes

One of the main findings of this research shows that unregulated traditional extensive livestock systems contribute strongly to the degradation of forest landscapes, pastoral lands and rural ecosystems in the Republic of Guinea. This dynamic is particularly visible in the agro-pastoral areas studied of Kindia, Labé, Mamou and Nzérékoré where pressures on natural resources have intensified over the past two (2) years under the combined effect of population growth, the expansion of agricultural activities, uncontrolled pastoral mobility, the inadequacy of land governance mechanisms and the growing effects of climate change.

Field surveys have shown that extensive pastoral practices continue to be dominated by free grazing, unplanned seasonal transhumance, pastoral bush fires, unregulated logging and the lack of organized grazing rotation.

These practices are gradually contributing to soil degradation, the disappearance of natural pastures, the reduction of vegetation cover, the fragmentation of ecosystems, the loss of biodiversity and the acceleration of land use conflicts.

6.1.1. Main forms of ecological degradation identified

Data collected from local communities, pastoralists, environmental NGOs and technical services have identified several major degradation factors linked to extensive livestock systems.

Table 17: Identified forms of ecological degradation

Main environmental pressures observed	Observed level of occurrence
Overgrazing of pastoral lands	Very high
Recurrent bushfires	High
Deforestation for pasture opening	High
Anarchic occupation of pastoral corridors	Moderate to High
Deterioration of pastoral water points	Moderate
Reduction of gallery forests	High

The participants particularly insisted on the increase in overgrazing around agricultural areas, classified forests, transhumance areas and ecological restoration perimeters.

According to several testimonies collected during the focus groups in Labé and Mamou, the gradual reduction of available space is now pushing herds to exploit ecologically fragile areas, accentuating the degradation of soil and plant resources.

6.1.2. Statistical results from quantitative surveys

The quantitative data collected through the questionnaires show a strong convergence of perceptions on the environmental impacts of extensive pastoral systems.

Table 18: Perception of the environmental impacts of extensive livestock farming

Identified environmental impacts	Percentage of respondents
Overgrazing is a major cause of land degradation	80 %
Extensive livestock farming increases pressure on forest resources	73 %
Bushfires worsen ecological degradation	67 %
Uncontrolled transhumance promotes land conflicts	71 %
The disappearance of natural pastures is accelerating	64 %
Pastoral water resources are gradually deteriorating	58 %

These results reveal that extensive pastoral systems are now one of the main factors of pressure on rural and forest landscapes in the areas studied.

6.1.3. Analysis of the interactions between extensive livestock farming and the SDGs

The analysis of the interactions between the SDGs shows that extensive livestock practices simultaneously produce several negative effects on SDG 13 (Climate Action), SDG 15 (Life on land), and indirectly SDG 2 (Food Security).

Table 19: Effects of extensive pastoral practices on the SDGs

Observed practices	Impacts on SDG 13	Impacts on SDG 15	Impacts on SDG 2
Overgrazing	Land degradation and reduced climate resilience	Ecosystem degradation	Decline in pastoral productivity
Bushfires	Carbon emissions	Loss of biodiversity	Reduction of fodder resources
Uncontrolled transhumance	Increased climate vulnerability	Fragmented habitat	Resource access conflicts
Pastoral deforestation	Reduced carbon sequestration capacity	Deforestation	Weakening livelihoods

This analysis highlights the existence of important trade-offs between the search for short-term pastoral productivity, the conservation of natural resources and the climate resilience of rural territories.

6.1.4. Landscape degradation and land conflicts

Beyond the ecological impacts, the study showed that increasing pressure on land is accentuating conflicts between farmers, herders, local communities and forest restoration projects.

The results of the questionnaires show that 70% of participants consider that land conflicts are increasing in rural areas, 62% believe that the lack of pastoral planning promotes territorial tensions and 59% consider that some forest restoration areas temporarily limit access to pastures.

Table 20: Main causes of natural resource conflicts

Causes of conflict	Percentage
Occupation of pastoral corridors	74 %
Competition between agriculture and livestock farming	71 %
Restriction of pastoral areas	63 %
Lack of local land mediation	57 %
Population pressure on land	66 %

Collective discussions have shown that conflicts become particularly frequent during the dry season when fodder resources are sharply depleted.

6.1.5. Observed impacts on forest landscapes

The participants interviewed also observed several major ecological transformations in the landscapes studied.

Table 21: Describes the main environmental transformations observed

Observed transformations	Frequency of occurrence in interviews
Reduction of vegetation cover	82 %
Degradation of natural pastures	76 %
Soil erosion	71 %
Decrease in water resources	65 %
Regression of gallery forests	61 %
Disappearance of certain plant species	54

Rural communities particularly highlighted the decrease in fodder trees, soil impoverishment, increased local droughts and the gradual scarcity of certain plant species used for animal feed.

6.1.6. Analytical summary of result 1

Overall, this first result shows that extensive livestock farming practices constitute a structural factor of ecological degradation and territorial vulnerability in Guinea when they are not accompanied by sustainable management mechanisms.

The results highlighted strong pressure on forest landscapes, progressive degradation of pastoral resources, increased land conflicts, weakening of ecosystem services and increased vulnerability of rural communities to climate change.

These findings demonstrate the need to strengthen pastoral governance systems, to supervise transhumance, to promote climate-smart agro-sylvo-pastoral practices and to develop local mechanisms for integrated natural resource management.

Thus, traditional pastoral practices appear not only as a major ecological issue, but also as a central issue of territorial governance and sustainability of socio-ecological systems in the Republic of Guinea.

However, beyond the negative impacts observed in extensive pastoral systems, the results of this research also show that several climate-smart agro-sylvo-pastoral practices tested locally offer sustainable alternatives capable of reducing environmental pressures while improving the living conditions of rural populations.

6.2. Major Result 2: Climate-smart agro-sylvo-pastoral practices generate positive synergies between food security, climate resilience and forest restoration

The second major result of this study highlighted the existence of important synergies between climate-smart agro-sylvo-pastoral practices, the restoration of forest landscapes and the improvement of the socio-economic resilience of rural communities.

The data collected showed that several local initiatives already implemented in some rural areas of Guinea are simultaneously contributing to the restoration of degraded ecosystems, improving agricultural and pastoral productivity, adapting to climate change, reducing pressure on natural forests, and strengthening the livelihoods of rural households.

These results demonstrate that integrated agriculture-livestock-forestry approaches represent strategic solutions capable of producing environmental, climate and socio-economic co-benefits while promoting the simultaneous achievement of SDGs 2 (Zero Hunger), SDG 13 (Climate Action) and SDG 15 (Life on Land).

6.2.1. Key climate-smart practices identified

The field consultations identified several agro-sylvo-pastoral practices considered by local actors to be particularly effective in strengthening the sustainability of rural systems.

Table 22: Key climate-smart practices identified

Observed practices	Main objectives	Observed effects
Agroforestry	Integration of trees, crops and livestock	Soil fertility, animal feed, reduction of deforestation
Assisted Natural Regeneration (ANR)	Natural plant restoration	Carbon sequestration, drilling, soil moisture
Rotational pastures	Sustainable rangeland management	Reduction of overgrazing, soil restoration
Silvopastoral systems	Tree-livestock association	Shading, animal feed, climate resilience
Improved forage crops	Strengthening animal feed	Reduction of pressure on natural pastures
Fodder reforestation	Pastoral biomass production	Reduction, uncontrolled logging and forest degradation

Respondents placed particular emphasis on the growing role of fodder trees, agroforestry hedgerows, local plant restoration techniques and integrated crop-livestock systems.

6.2.2. Statistical results of climate-smart practices

Quantitative surveys show a strong recognition of the benefits associated with integrated agro-sylvo-pastoral practices.

Table 23: Perception of benefits of climate-smart practices

Practices and perceived benefits	Percentage of respondents
Agroforestry reduces pressure on forests	74 %
Agroforestry systems improve climate resilience	71 %
FMNR improves soil fertility	69 %
Silvopastoral systems improve animal feed	66 %
Pasture rotations reduce land degradation	63 %
Forage trees improve the availability of fodder in the dry season	68 %
Integrated practices boost rural incomes	59

These results reveal a strong convergence of perceptions between rural producers, pastoralists, NGOs, researchers and public institutions.

6.2.3. Contribution of climate-smart practices to the SDGs

The analysis of the interactions between the SDGs highlights that agro-sylvo-pastoral practices generate cross-benefits between food security, climate resilience and ecological restoration.

Table 24: Contributions of climate-smart practices to the SDGs

Practices	Contribution to SDG 2	Contribution to SDG 13	Contribution to SDG 15
Agroforestry	Increased agricultural and forage productivity	Climate adaptation	Landscape restoration
NAS	Securing forage	Carbon sequestration	Ecosystem regeneration
Silvopastoral systems	Improved animal feed	Reducing climate vulnerability	Biodiversity Conservation
Rotational pastures	Sustainable pastoral productivity	Reduced soil degradation	Protection of natural pastures
Crop drilling	Reduction of pastoral food insecurity	Drought adaptation	Reduction of forest pressure

This analysis confirms that integrated approaches can reduce the trade-offs traditionally observed between pastoral production, forest conservation and climate adaptation.

6.2.4. Assisted natural regeneration (ANR) as a strategic solution

One of the most important findings of this research concerns the central role of assisted natural regeneration (FMNR) in the restoration of agro-pastoral landscapes. The communities interviewed indicated that FMNR allowed the gradual return of certain plant species, the restoration of degraded soils, the improvement of land moisture, the availability of shade for livestock and the increase of fodder resources during the dry season.

Table 25: Main benefits of NAS observed

Benefits of FMNR	Percentage of occurrence in interviews
Improved forage	71 %
Improved soil fertility	68 %
Erosion Reduction	64 %
Improved soil moisture	61 %
Carbon sequestration	57 %
Regeneration of local species	59 %

The focus groups carried out in Kindia and Mamou showed that the FMNR is a solution particularly suited to rural areas with limited financial means, as it is largely based on the development of the natural capacity for ecosystem regeneration.

6.2.5. Socio-economic effects of integrated practices

The study also paid particular attention to climate-smart practices that produce several indirect benefits for rural communities.

Participants indicated that these practices contribute to the diversification of incomes, the improvement of animal feed, the reduction of expenses related to livestock feed, the security of agricultural production and the improvement of the economic resilience of households.

Table 26: Main socio-economic benefits observed

Socio-economic benefits	Percentage of respondents
Reduction of drought-related agricultural losses	66 %
Improving rural incomes	59 %
Securing animal feed	68 %
Reduced resource conflicts	54 %
Diversification of rural economic activities	57

The women interviewed particularly emphasized the role of fruit trees, the exploitation of non-timber forest products and the economic opportunities linked to agroforestry systems.

6.2.6. Observed Limitations and Constraints

Despite their benefits, several constraints still limit the widespread adoption of climate-smart practices.

The main difficulties mentioned in this study concern the lack of technical support, difficulties in accessing financing, the inadequacy of extension policies, land conflicts and weak institutional coordination.

Table 27: Constraints to the adoption of climate-smart practices

Identified constraints	Percentage
Lack of technical support	72 %
Difficulties in accessing finance	69 %
Weak institutional coordination	63 %
Land insecurity	58 %
Insufficient local training	61 %

These results show that agro-sylvo-pastoral innovations require more structured institutional support in order to guarantee their sustainability and dissemination.

6.2.7. Analytical synthesis of result 2

Overall, this second result demonstrates that climate-smart agro-sylvo-pastoral practices are strategic levers capable of reconciling forest restoration, food security, climate resilience and improved livelihoods.

The results show that integrated approaches can restore the ecological functions of landscapes, reduce pressure on forests, improve the productivity of rural systems and strengthen the resilience of communities to climate change.

The research has shown that the synergies between SDGs 2, 13 and 15 become truly operational when environmental policies integrate pastoral realities, local economic needs, territorial governance mechanisms and community knowledge.

These results confirm that integrated agro-sylvo-pastoral approaches represent one of the most promising options for a sustainable ecological and climate transition in the Republic of Guinea.

Despite the progress observed through certain climate-smart agro-sylvo-pastoral practices, the results of this research show that their scaling up remains severely limited by institutional inadequacies, territorial governance problems and weak coordination between sectoral actors.

In particular, the study reveals that environmental, agricultural, pastoral and land policies continue to be implemented in a fragmented manner, which significantly reduces the effectiveness of forest restoration and sustainable livestock initiatives. These institutional limitations are now one of the main obstacles to operationalizing synergies between SDGs 2, 13 and 15.

6.3. Major Result 3: Governance and institutional coordination deficiencies limit the integration between forest restoration and sustainable livestock

This result of this research highlights the central role of territorial governance in the success or failure of integrated forest restoration and sustainable livestock initiatives in the Republic of Guinea.

Qualitative and quantitative analyses show that the difficulties of integration between environmental policies, agricultural policies, pastoral strategies and land tenure mechanisms constitute a major obstacle to the sustainability of rural landscapes. The target actors interviewed stressed that the lack of institutional coordination promotes the duplication of interventions, conflicts over land use, land insecurity, low community ownership of projects and the ineffectiveness of local natural resource management mechanisms.

6.3.1. Weak coordination between sectoral policies

The study also showed that public policies relating to forest restoration, livestock, agriculture and land management are often developed and implemented in a sectoral manner without any real integrated territorial planning mechanism.

The institutional actors interviewed indicated that environmental strategies do not sufficiently take into account pastoral realities, agricultural policies do not integrate ecological restoration issues to a very limited extent, land tenure systems remain poorly adapted to pastoral mobility and local consultation frameworks remain limited.

Table 28: Main institutional shortcomings identified

Institutional constraints observed	Level of occurrence
Weak intersectoral coordination	Very high
Insufficient consultation mechanisms	High
Low community involvement	High
Lack of pastoral planning	High
Land management difficulties	Very high
Insufficient technical extension	Moderate to High

The results thus reveal a significant fragmentation of public interventions in the rural areas studied.

6.3.2. Statistical results on environmental governance

The quantitative data obtained through the questionnaires confirm the widespread perception of institutional weakness in the management of the interactions between forest restoration and sustainable livestock farming.

Table 29: Perception of territorial governance challenges

Institutional challenges identified	Percentage of respondents
Restoration policies suffer from a lack of intersectoral coordination	72 %
Land conflicts are on the rise in rural areas	70 %
Participatory approaches need to be strengthened	74 %
Local mediation mechanisms are insufficient	67 %
Local communities are insufficiently involved	69 %
Pastoral policies remain poorly operational	63 %

These results show that governance issues now appear to be a central dimension of ecological restoration and climate resilience processes.

6.3.3. Land conflicts as the main factor of territorial tension

The study reveals that the growing competition for natural resources is increasing tensions between farmers, herders, local communities, forest restoration projects and local authorities.

The study areas of Kindia, Labé and Nzérékoré particularly highlighted the increase in conflicts related to transhumance corridors, the occupation of agricultural land, the temporary exclusion of restored areas and the boundaries of protected forest areas.

Table 30: Main Causes of Land Disputes

Causes of conflict	Percentage
Occupation of pastoral corridors	74 %
Crop-livestock competition	71 %
Restrictions on restoration projects	63 %
Land insecurity	68 %
Lack of local mediation	57 %
Population pressure on land	66

Interviewees explained that conflicts become particularly common during the dry season when grazing land decreases, water resources become scarce and agricultural land expands.

6.3.4. The social trade-offs of forest restoration policies

The analysis of the interactions between the SDGs also showed that some restoration policies can produce social trade-offs when they do not sufficiently take into account the needs of rural communities.

Several pastoralists interviewed said that some restored areas are becoming temporarily inaccessible to grazing, forest restrictions sometimes limit the movement of livestock, and conservation measures are sometimes perceived as economic constraints.

These results show that restoration initiatives can create tensions between SDG 15 (ecosystem restoration) and SDG 2 (food security and pastoralism).

Table 31: Main trade-offs identified between catering and livestock farming

Trade-offs observed	Potential effects
Restriction of grazing space	Tensions with farmers
Defending restored areas	Temporary reduction in forage
Strict forest protection	Limitation of pastoral activities
Intensification of livestock farming	Exclusion of poor households
Cost of climate innovations	Inequalities in access to sustainable practices

This analysis confirms that the success of environmental policies depends heavily on their ability to integrate local socio-economic realities.

6.3.5. The economic limits of semi-intensive livestock farming

The study also shows that semi-intensive livestock systems, although more ecologically sustainable, remain difficult to access for vulnerable rural households.

The main constraints mentioned concern the high cost of investments, difficulties in accessing financing, the lack of technical training and the inadequacy of pastoral infrastructure.

Table 32: Constraints to the adoption of semi-intensive systems

Identified constraints	Percentage
Difficulty accessing financing	73 %
High cost of investment	70 %
Lack of technical support	66 %
Poor access to pastoral inputs	59 %
Lack of rural infrastructure	61 %

These results show that climate-smart innovations are likely to benefit households with greater economic resources most, which can exacerbate some rural inequalities.

6.3.6. Need for integrated territorial governance mechanisms

Faced with these difficulties, the participants largely insisted on the need to strengthen multi-stakeholder platforms, local land mediation mechanisms, participatory governance, pastoral planning and integrated agriculture-livestock-forestry approaches.

Table 33: Priority Solutions Proposed by Participants

Proposed solutions	Percentage
Strengthening participatory approaches	74 %
Setting up multi-stakeholder platforms	69 %
Development of land mediation mechanisms	66 %
Strengthening territorial planning	63 %
Technical support to rural communities	71

The results confirm that local communities want to be more involved in the management of natural resources, in the definition of environmental policies and in territorial governance mechanisms.

6.3.7. Analytical synthesis of result 3

Overall, this third result demonstrates that inadequacies in governance and institutional coordination are now one of the main obstacles to the sustainable integration of forest restoration and climate-smart livestock in Guinea. The study highlighted the fragmentation of sectoral policies, a weakness in consultation mechanisms, an increase in land conflicts, social compromises related to restoration policies and inequalities in access to sustainable innovations.

The results highlighted that the transition to resilient agro-sylvo-pastoral systems does not depend only on technical innovations, but also on the quality of institutions, territorial governance, community participation, land tenure security and the effective integration of public policies.

This research has finally shown that the synergies between SDGs 2, 13 and 15 cannot be sustainable without the establishment of inclusive environmental governance mechanisms capable of reconciling ecosystem conservation, food security, social justice and climate resilience.

Beyond the institutional constraints and governance challenges identified above, the results of this research also highlighted that integrated approaches to forest restoration and sustainable livestock generation generate several indirect benefits for rural communities, production systems and local ecosystems.

In particular, the study reveals that agroforestry, silvopastoral systems and assisted natural regeneration practices not only contribute to environmental restoration, but also contribute to the diversification of livelihoods, the strengthening of climate resilience, the improvement of rural incomes, the restoration of ecosystem services and the progressive conservation of local biodiversity.

These results highlighted the potential of integrated approaches as a lever for sustainable rural development and inclusive ecological transition.

6.4. Outcome 4: Integrated approaches generate collateral benefits for livelihoods and biodiversity

The fourth result of this research highlights the multiple socio-economic, ecological and climatic benefits generated by integrated forest restoration and sustainable livestock farming practices in the localities studied.

Qualitative and quantitative analyses have shown that agroforestry and silvopastoral systems simultaneously improve the living conditions of rural households, strengthen food and pastoral security, reduce climate vulnerability, restore certain ecological functions of landscapes and promote the gradual regeneration of local natural resources.

The results also showed that participatory approaches, strengthen community ownership of environmental projects and improve the sustainability of local restoration initiatives.

6.4.1. Contribution of integrated approaches to rural livelihoods

The communities interviewed largely emphasized that agro-sylvo-pastoral systems contribute to the diversification of income and the economic security of rural households.

The main economic benefits observed concern the improvement of animal feed, the increase in certain agricultural productions, the increased availability of fodder, the development of non-timber forest products and the income generated by fruit and fodder trees.

Table 34: Main socio-economic benefits identified.

Observed benefits	Percentage of respondents
Improving animal feed	72 %
Diversification of rural incomes	67 %
Reduction of drought-related agricultural losses	64 %
Improved food security	69 %
Valorization of non-timber forest products	61 %
Reduced fodder expenses	58%

These results show that integrated approaches can gradually reduce the economic vulnerability of rural households to climatic hazards, droughts and the degradation of natural resources.

6.4.2. Role of women and community groups

The study also reveals a significant involvement of women and community-based organizations in local ecological restoration initiatives.

The women interviewed particularly emphasized the economic importance of non-timber forest products, the benefits of fruit trees, opportunities for local processing, securing water resources and improving agricultural soils.

Table 35: Main benefits identified by women and community groups

Benefits mentioned	Frequency of occurrence
Income from non-timber forest products	68 %
Improvement of water resources	63 %
Increased fertility of agricultural soils	66 %
Availability of fruit trees	59 %
Reduction of timber collection distances	52 %
Improving family nutrition	60%

The focus groups organized in Kindia and Mamou showed that women often play a central role in the management of nurseries, the protection of restored areas and the enhancement of local forest resources.

6.4.3. Effects of integrated practices on biodiversity and landscapes

One of the important results of this study concerns the positive effects observed on local biodiversity and the ecological functions of the restored landscapes.

Participants indicated that areas benefiting from agroforestry practices or assisted natural regeneration show an improvement in vegetation cover, a gradual return of certain local species, a reduction in soil erosion, an improvement in land moisture and a better regulation of water resources.

Table 36: Key Ecological Effects Observed

Observed ecological effects	Percentage of respondents
Regeneration of vegetation cover	74 %
Reduced soil erosion	69 %
Return of some local plant species	63 %
Improved soil moisture	61 %
Stabilization of degraded land	58 %
Improvement of water resources	57 %

Field observations show that restored areas are gradually helping to reconnect certain ecological habitats, improve soil biological fertility and restore the ecological functions of rural landscapes.

6.4.4. Contribution of integrated practices to climate resilience

The results also show that integrated approaches significantly improve the climate resilience of rural production systems.

Participants explained that trees integrated into agricultural and pastoral systems help reduce the effects of high heat, maintain soil moisture, improve crop resilience to drought and secure fodder resources during critical periods.

Table 37: Impacts of Integrated Practices on Climate Resilience

Observed climate effects	Percentage
Reduced vulnerability to drought	71 %
Maintaining soil moisture	64 %
Increased availability of forage in the dry season	68 %
Reducing land degradation	66 %
Improving the resilience of rural farms	62%

These results show that integrated practices are particularly important local adaptation strategies in a context marked by erratic rainfall, recurrent droughts and progressive land degradation.

6.4.5. Participatory approaches as a factor for sustainability

The study also shows that projects that actively involve local communities tend to have better ownership, greater sustainability and reduced conflicts over natural resources.

Participants stressed that participatory approaches promote acceptance of restoration measures, community protection of restored areas, and collective management of natural resources.

Table 38: Effects of participatory approaches on local governance

Observed effects	Percentage
Better ownership of projects	73 %
Reduction of local conflicts	59 %
Strengthening Community Governance	64 %
Increased participation of local populations	69 %
Community protection of restored spaces	66 %

These results demonstrate that community participation is a key factor in the success of ecological restoration and sustainable livestock initiatives.

Analysis of synergies between SDGs 2, 13 and 15

Analysis of the interactions between the SDGs shows that integrated approaches produce several simultaneous positive synergies.

Table 39: Synergies between the SDGs

Integrated practices	Contribution to SDG 2	Contribution SDG 13	Contribution SDG 15
Agroforestry	Food safety	Climate adaptation	Ecological restoration
NAS	Drilling availability	Carbon sequestration	Regeneration Plant
Silvopastoral systems	Pastoral productivity	Climate resilience	Biodiversity Conservation
Fodder reforestation	Animal feed	Reduced vulnerability	Reduction of forest pressure

This analysis confirms that integrated approaches can transform potential trade-offs between production and conservation into real territorial synergies.

6.4.6. Analytical synthesis of the result 4

Overall, this fourth result demonstrates that integrated forest restoration and sustainable livestock practices produce multiple benefits that go far beyond the environmental framework alone.

The results highlight a gradual improvement in livelihoods, diversification of rural incomes, strengthening of climate resilience, restoration of ecological functions of landscapes, and better conservation of some local natural resources.

The study also shows that participatory approaches promote community ownership, local governance and the sustainability of environmental projects, thus, integrated agro-sylvo-pastoral practices are not only ecological solutions, but also strategic levers for sustainable rural development, social inclusion and territorial resilience in the Republic of Guinea.

Finally, the regional comparative analysis carried out as part of this research identified several strategic lessons that could strengthen national policies on forest restoration, environmental governance and sustainable livestock farming in the Republic of Guinea. The study and desk research show that several West African countries facing similar challenges of land degradation, pastoral pressure and climate vulnerability have gradually developed integrated territorial approaches based on community participation, concerted management of natural resources, agriculture-livestock-forestry integration, land tenure security and multi-stakeholder governance.

These regional experiences offer important references for improving Guinea's national climate resilience and forest landscape restoration strategies.

6.5. Result 5: Regional experiences show the importance of integrated and participatory territorial approaches

The fifth major result of this research stems from the comparative analysis of forest restoration and pastoral management experiences observed in several West African countries engaged in: the Bonn Challenge, the AFR100 initiative, Land Degradation Neutrality (LDN) policies and national climate adaptation strategies.

Benchmarking conducted at:

- Burkina Faso;
- Senegal;
- Liberia;
- Côte d'Ivoire.

Show that the most sustainable initiatives are mainly based on the active participation of local communities, integrated territorial governance, land tenure security, intersectoral coordination and the integration of agro-sylvo-pastoral systems.

The results also show that approaches that focus exclusively on technical reforestation without taking into account local socio-economic needs often produce limited long-term results.

6.5.1. Key lessons learned from regional experiences

The benchmarking study identified several success factors common to the most successful regional initiatives.

Table 40: Key success factors observed in regional experiences

Success factors	Observed level of importance
Community participation	Very high
Integrated land management	Very high
Land tenure security	High
Intersectoral coordination	High
Agro-sylvo-pastoral integration	Very high
Technical support for producers	Moderate to High
Participatory local governance	High

The analysis shows that programmes that simultaneously integrate ecological conservation, food security, local governance, rural development and tend to achieve more sustainable results that are better accepted by the population.

6.5.2. Comparative analysis of regional approaches

The study reveals significant differences between the restoration models implemented in the countries studied.

Table 41: Comparison of Regional Approaches to Restoration

Countries studied	Main approaches observed	Main results
Burkina Faso	Assisted natural regeneration and community management	Soil restoration and degradation reduction
Senegal	Agroforestry and the Great Green Wall	Climate resilience and pastoral security
Liberia	Forest Community Governance	Reducing forest conflicts
Ivory Coast	Cocoa-forest agroforestry systems	Diversification of rural incomes

Experiences show that approaches based on community involvement, resource co-management and agroforestry systems generally have better results in terms of ecological and socio-economic sustainability.

6.5.3. Limitations of Reforestation-Only Approaches

The comparative analysis also shows that several regional initiatives based exclusively on technical reforestation have encountered significant limitations.

The main difficulties identified concern the low level of community ownership, the inadequacy of monitoring mechanisms, conflicts over land use, the lack of consideration of pastoral needs and the high costs of maintenance.

Table 42: Main limitations of conventional reforestation approaches

Observed constraints	Percentage of occurrence in the comparative studies
Low community involvement	71 %
Difficulties in maintaining plantations	67 %
Land conflicts	63 %
Poor consideration of pastoral needs	66 %
Dependence on external financing	59 %

These results show that restoration projects that ignore local socio-economic realities often have low long-term sustainability.

6.5.4. Contribution of integrated approaches to conflict reduction and climate resilience

Comparative analyses reveal that integrated territorial approaches can produce several simultaneous benefits.

Table 43: Observed Effects of Integrated Territorial Approaches

Observed effects	Percentage of occurrence
Reducing land conflicts	68 %
Improving rural incomes	65 %
Building climate resilience	72 %
Stabilization of degraded landscapes	69 %
Enhanced community participation	74 %
Better territorial governance	66

The results show that integrated agro-sylvo-pastoral systems allow for better use of natural resources, reduced pressures on forests, improved food security and greater resilience of rural communities to climate change.

6.5.5. Policy lessons for Guinea

The regional experiences studied make it possible to identify several strategic orientations that are particularly relevant to the Guinean context.

Table 44: Key strategic orientations tailored to Guinea

Proposed Strategic Directions	Objectives
Strengthening local governance	Conflict Reduction and Community Participation
Development of agro-sylvo-pastoral systems	Integration of SDGs 2, 13 and 15
Securing pastoral land tenure	Reduction of territorial tensions
Promotion of the NAS	Low-cost ecological restoration
Strengthening multi-stakeholder platforms	Institutional coordination
Support for local innovations	Climate resilience and social inclusion

These guidelines demonstrate that the success of environmental policies in Guinea depends heavily on the integration of local actors, territorial planning, institutional coherence and participatory governance.

6.5.6. Analysis of regional synergies between SDGs 2, 13 and 15

The benchmarking also confirms that integrated approaches can simultaneously strengthen several Sustainable Development Goals.

Table 45: Regional synergies among the SDGs

Integrated approaches	SDG 2	SDG 13	SDG 15
Agroforestry	Food safety	Climate adaptation	Forest restoration
NAS	Drilling availability	Carbon sequestration	Landscape regeneration
Community Management	Rural income	Territorial resilience	Biodiversity Conservation
Silvopastoral systems	Pastoral productivity	Reduced vulnerability	Soil protection

This analysis shows that integrated territorial approaches can transform potential trade-offs between production, conservation and climate adaptation into real synergies for sustainable development.

Overall analytical conclusion of the results

Overall, the results of this research showed that forest landscape restoration and climate-smart livestock practices can produce important synergies between food security, climate resilience, ecosystem conservation and sustainable rural development.

The study particularly highlighted the role of agro-sylvo-pastoral practices in reducing environmental pressures, the socio-economic benefits of integrated approaches, the importance of participatory territorial governance, the need to manage land and institutional trade-offs, and the importance of community knowledge in ecological restoration processes.

The analyses also showed that the synergies between SDG 2 (Zero Hunger), SDG 13 (Climate Action) and SDG 15 (Life on Land) remain strongly conditioned by the quality of governance mechanisms, the coordination of public policies, the participation of local communities, the socio-economic inclusion of rural populations and the security of land resources.

Table 46: Key Analytical Research Findings

Main results obtained	Strategic implications
High pressure from extensive systems on landscapes	Need for agro-sylvo-pastoral transition
High potential for climate-smart practices	Development of integrated approaches
Weakness of territorial governance	Strengthening participatory mechanisms
Importance of socio-economic benefits	Integration of rural development and conservation
Relevance of regional experiences	Adaptation of good practices to the Guinean context

Ultimately, this research demonstrated that the success of forest restoration and climate resilience strategies in the Republic of Guinea depends less on technical interventions alone than on the ability of public, community and institutional actors to build integrated, inclusive, resilient and socially equitable territorial models.

7. Conclusion

This research analyzed the interactions between forest landscape restoration and climate-smart livestock practices in the Republic of Guinea in a context marked by land degradation, climate change and increasing pressure on natural resources. The results obtained showed that uncontrolled extensive livestock systems contributed significantly to ecological degradation, particularly through overgrazing, bush fires and land-use conflicts.

However, the study also demonstrated that several integrated practices, such as agroforestry, assisted natural regeneration and silvopastoral systems, produced positive synergies between food security, climate resilience and ecosystem restoration. The analyses highlighted that the success of these approaches depends strongly on the quality of territorial governance, community participation and the integration of environmental, agricultural and pastoral policies.

Ultimately, this research has shown that integrated and participatory territorial approaches are strategic levers for reconciling sustainable rural development, natural resource conservation and climate adaptation in Guinea.

8. Recommendations

In view of the results obtained, several strategic recommendations appear necessary to strengthen the synergies between forest restoration, sustainable livestock and climate resilience in the Republic of Guinea.

1. Strengthen the integration of environmental, agricultural, pastoral and land policies into national sustainable development strategies;
2. Develop local participatory governance mechanisms involving rural communities, pastoralists, local authorities and civil society organizations;
3. Promote climate-smart agro-sylvo-pastoral practices, including agroforestry, assisted natural regeneration and sustainable pasture management;
4. Secure transhumance corridors and improve land management to reduce conflicts between farmers and herders;
5. Strengthen access to climate finance, technical training and innovations for vulnerable rural households;
6. Establish multi-stakeholder platforms to improve intersectoral coordination and monitoring of restoration initiatives;
7. Promote local knowledge and community experiences in ecological restoration and climate adaptation policies;
8. Strengthen awareness programmes on the environmental impacts of overgrazing and bush fires;
9. Develop monitoring and evaluation systems based on reliable scientific and territorial data;
10. Encourage regional cooperation and sharing of experiences with West African countries engaged in forest landscape restoration initiatives.

9. Key references

Republic of Guinea – Ministry of Environment and Sustainable Development (MEDD). Guinea's First Biennial Update Report on Climate Change. Conakry, Guinea, 2024, 214 p.

Republic of Guinea – Ministry of Environment and Sustainable Development (MEDD). Restoration Opportunity Assessment Mechanism (ROAM) in Guinea. Conakry, Guinea, 2021, 167 p.

Republic of Guinea – Ministry of Environment and Sustainable Development.
National Forest Restoration Plan (PNRF). Conakry, Guinea, 2022, 132 p.

Food and Agriculture Organization of the United Nations (FAO). Review of Forest and Landscape Restoration in Africa. Rome, Italy, 2020, pp. 45-89.

FAO. Agroforestry systems for dryland resilience. Rome, Italy, 2022, 96 p.

United Nations Development Programme (UNDP Guinea). Report on sustainable agro-sylvo-pastoral systems and climate resilience in Guinea. Conakry, Guinea, 2021, 118 p.

United Nations Development Programme (UNDP Guinea). Climate Resilience and Landscape Restoration Project in Dabola Prefecture. Conakry, Guinea, 2023, 74 p.

Forestry Investment Fund (FIP Guinea). Study on assisted natural regeneration and community restoration of degraded land in Guinea. Conakry, Guinea, 2023, 81 p.

ECOWAS / CILSS. PRAPS: Regional Support Project for Pastoralism in the Sahel – Pastoral Resilience and Natural Resource Governance. Ouagadougou, Burkina Faso, 2021, pp. 32-76.

International Union for Conservation of Nature (IUCN) & FAO. FLR Africa – Guinea Project: Integrated Approaches to Forest Landscape Restoration. Gland, Switzerland
/ Rome, Italy, 2022, 102 p.

United Nations Convention to Combat Desertification (UNCCD). Global Land Outlook – Land Restoration and Resilience. Bonn, Germany, 2022, pp. 115-154.

Donzo, V., Field surveys on the synergies between forest restoration and climate-smart livestock farming in Guinea. Individual interviews conducted in Conakry, Kindia, Labé and Nzérékoré (1–28 August 2025); Focus groups organized in Conakry, Kindia, Mamou and Kankan (05–14 September 2025); Questionnaire surveys conducted in Conakry, Dubréka, Kindia and Labé (02–22 October 2025). Primary Research Data, 2025.